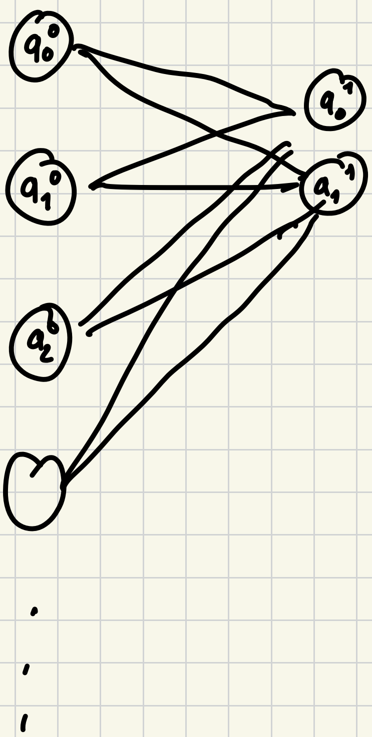
$\sigma : \mathbb{R} \to (-1, 1)$

$x \mapsto \frac{1}{1+e^{-x}}$

0th layer activations

1st Layer activations

$$\sigma\left(\begin{bmatrix} w_{0,0} & w_{0,1} & \cdots & w_{0,n} \\ & & & \\ w_{k,0} & \cdots & & w_{k,n} \end{bmatrix}\begin{bmatrix} a_0^0 \\ a_1^0 \\ a_2^0 \\ \vdots \\ a_n^0 \end{bmatrix} + \begin{bmatrix} b_0^0 \\ b_1 \\ \vdots \\ b_k \end{bmatrix}\right) = \begin{bmatrix} a_0^1 \\ a_1^1 \\ \vdots \\ a_n^1 \end{bmatrix}$$



$$a_0^{(1)} = w_{00}\, a_0^0 + w_{01}\, a_1^0 \; -$$

$$a_1^1 = w_{10} a_0^0 + w_{11} a \cdots$$

$$\frac{d\,(1+e^{-x})^{-1}}{d(1+e^{-x})} \qquad \frac{d\; 1+e^{-x}}{dx}$$

$$= \frac{e^{-x}}{(1+e^{-x})^2} X$$

MNIST DB

Gradient Descent: way to converge to a local minima.

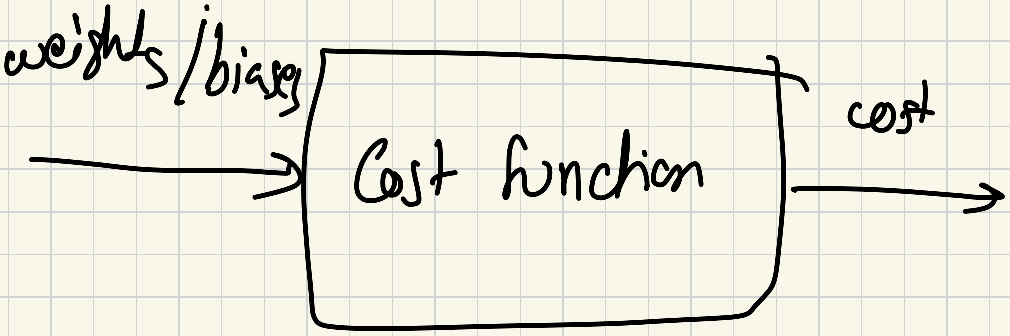
Training:

1) initialise randomly

(add up squares of differences of output and expected output)

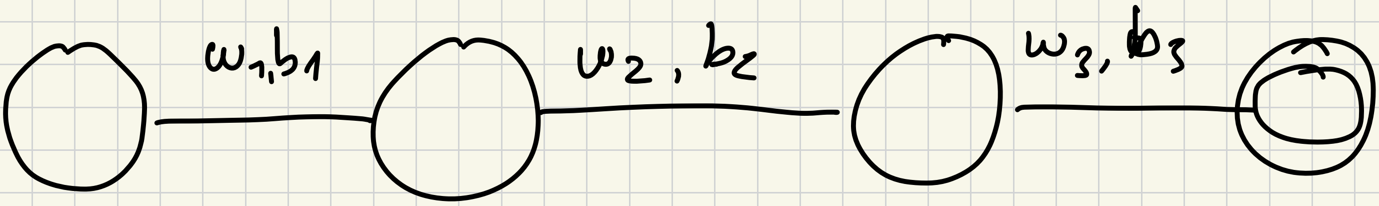
→ cost

2) Calculate Average Cost

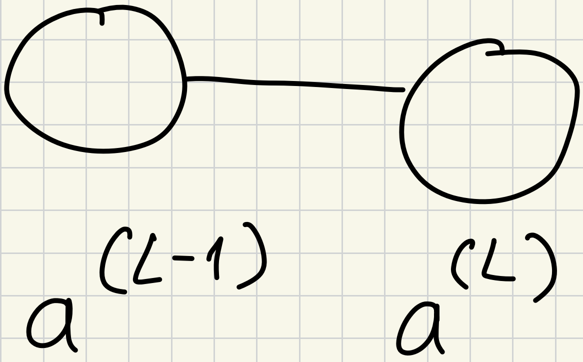


multivar $\nabla C(x,y)$ direction of steepest ascent at (x,y)

compute ∇C small step in $-\nabla C$ direction



$$C(w_1, b_1, w_2, b_2, w_3, b_3)$$



y: Desired output

Cost $\quad C_0 = (a^L - y)^2$

$$z^{(L)} = w^{(L)} a^{(L-1)} + b^{(L)}$$

$$a^{(L)} = \sigma(z^{(L)})$$

$$\frac{\partial C_0}{\partial w^{(L)}} = \frac{\partial z^{(L)}}{\partial w^{(L)}} \times \frac{\partial a^{(L)}}{\partial z^{(L)}} \times \frac{\partial C_0}{\partial a^{(L)}}$$

$$\frac{\partial C_0}{\partial a^{(L)}} = 2(a^{(L)} - y)$$

$$\frac{\partial a^{(L)}}{\partial z^{(L)}} = \sigma'(z^{(L)})$$

$$\frac{\partial z^{(L)}}{\partial w^{(L)}} = a^{(L-1)}$$